

Modulation of Phonetic Realizations in Cantonese Dialogue with Human and AI Interlocutors



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Introduction

- **Speech modulation**
 - Refers to speakers' **adaptive adjustments** to an interlocutor's characteristics.
 - Human-Human Interaction (HHI)**:
 - Widely documented across **different interaction types** [1,2,3] (e.g., child-, adult-, and foreigner-directed speech).
 - Human-AI Interaction (HAI)**:
 - When interacting with **prosocial/positive AI speech**, humans modulated speech rate and duration [4,5].
 - Little** is known about how humans respond to **negative or "antagonistic" AI speech**.
 - Fine-grained modulation** in **interactive dialogue** and **tone languages** remains underexplored.
- **Current study**: to investigate phonetic modulation in **Hong Kong Cantonese (HKC)** using **scripted dialogues**.
- **Research questions**
 - Does **interlocutor type (Human vs. AI)** influence **fine-grained acoustic realizations** in HKC?
 - Does **emotional context (Neutral vs. Negative)** further modulate these interlocutor-related effects?

Method

- **Participants**: 17 native HKC speakers (9M, 8F; 18-27 yrs)
 - **Experiment design**: **scripted** dialogue task (Wizard-of-Oz)
 - Stimuli**: one target x four times / dialogue
 - Within-subject factor 1: Interlocutor types**
- Comparable posture and facial expression; speaker-specific characteristics and prosody between human and AI interlocutors**
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10 dialogues with human interlocutor



10 dialogues with AI interlocutor

Audio-visual recordings of an HKC speaker

Animated Memoji avatar; specifically-customized AI voice
- Highlights of our AI voice:**
- Trained on **~4000 Cantonese sentences** recorded by **the same HKC speaker**
 - Using a **modified duration-informed DurlAN model** with phoneme duration, pitch, and energy as input
 - Closely reproduced** the original speaker's **prosody, rhythm, and pronunciation style**, while retaining a slightly mechanical quality

- **Within-subject factor 2: Emotional context (neutral vs. negative)**

Each dialogue consisted of **three** turn pairs:

Neutral opening turn (target elicitation) → Emotional context buildup → **Negative** context turn (target elicitation)

- **Data analysis**:
 - Speech rate & target word/syllable duration: Linear mixed-effects models (LMMs)
Interlocutor Type × Emotional Context
 - F0 trajectories of target syllables: Generalized additive mixed models (GAMMs)
Tone Category × Interlocutor Type × Emotional Context
 - Vowel formants: Vowel Space Area (VSA) computed from three corner vowels; linear mixed-effects models (LMMs)
Interlocutor Type × Emotional Context

References: [1] C. Cox et al., "A systematic review and Bayesian meta-analysis of the acoustic features of infant-directed speech," Nat. Hum. Behav., vol. 7, no. 1, pp. 114–133, Jan. 2023. [2] Y. Wang, A. Seidl, and A. Cristia, "Acoustic-phonetic differences between infant- and adult-directed speech: the role of stress and utterance position," J. Child Lang., vol. 42, no. 4, pp. 821–842, Jul. 2015. [3] A. Al-Kendi and G. Khattab, "Acoustic properties of foreigner directed speech," in Proc. 19th International Congress of Phonetic Sciences, 2019. Accessed: Jun. 18, 2026. [4] M. Cohn and G. Zellou, "Expressiveness Influences Human Vocal Alignment Toward voice-AI," in Interspeech, 2019, pp. 41–45. [5] M. Cohn, K. Predeck, M. Sarian, and G. Zellou, "Prosodic alignment toward emotionally expressive speech: Comparing human and Alexa model talkers," Speech Commun., vol. 135, pp. 66–75, Dec. 2021.

Results

- **Speech rate & Syllable/Word duration**
 - Utterance-level speech rate**: **no** significant differences between human and AI interlocutors or between neutral and negative emotions.
 - Syllable/word duration**: significant effect of **interlocutor**: **shorter in AI condition** (word: $\beta = -10.891$, $p = .04$; syllable: $\beta = -5.986$, $p = .06$). No effect of emotional context (Figure 1, left panel).

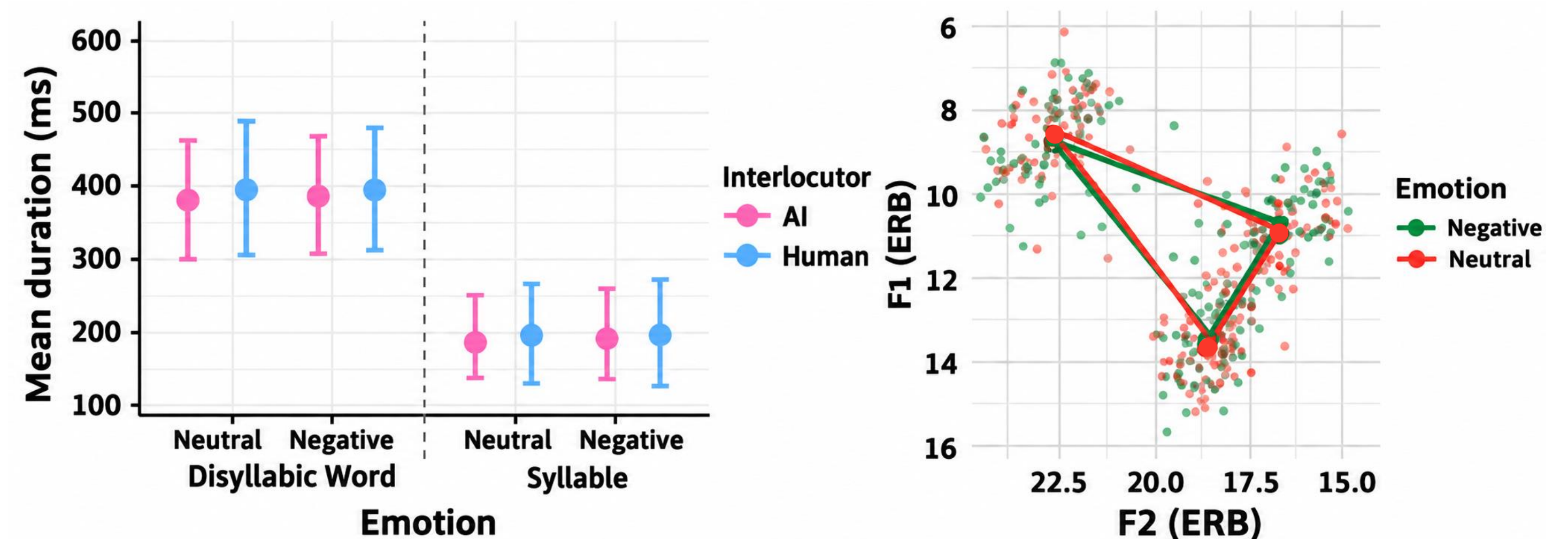


Figure 1: Mean duration of disyllabic target words and syllables (left; error bars = standard deviation) and vowel space area across emotional contexts (right).

- **Vowel space expansion**
Marginal effect of **emotion** on **vowel space area** ($\beta = 1.07$, $p = .057$), with **slightly larger VSA** in **negative** relative to neutral emotion (Figure 1, right panel). No effect of interlocutor type.
- **F0 trajectory**

Interlocutor and emotion effects were **tone-specific**.

Interlocutor effect: **AI > human**: T1 & T4 / neutral, T4 / negative;
AI < human: T5 / neutral

Emotion effect: both interlocutors: T2 **negative > neutral**

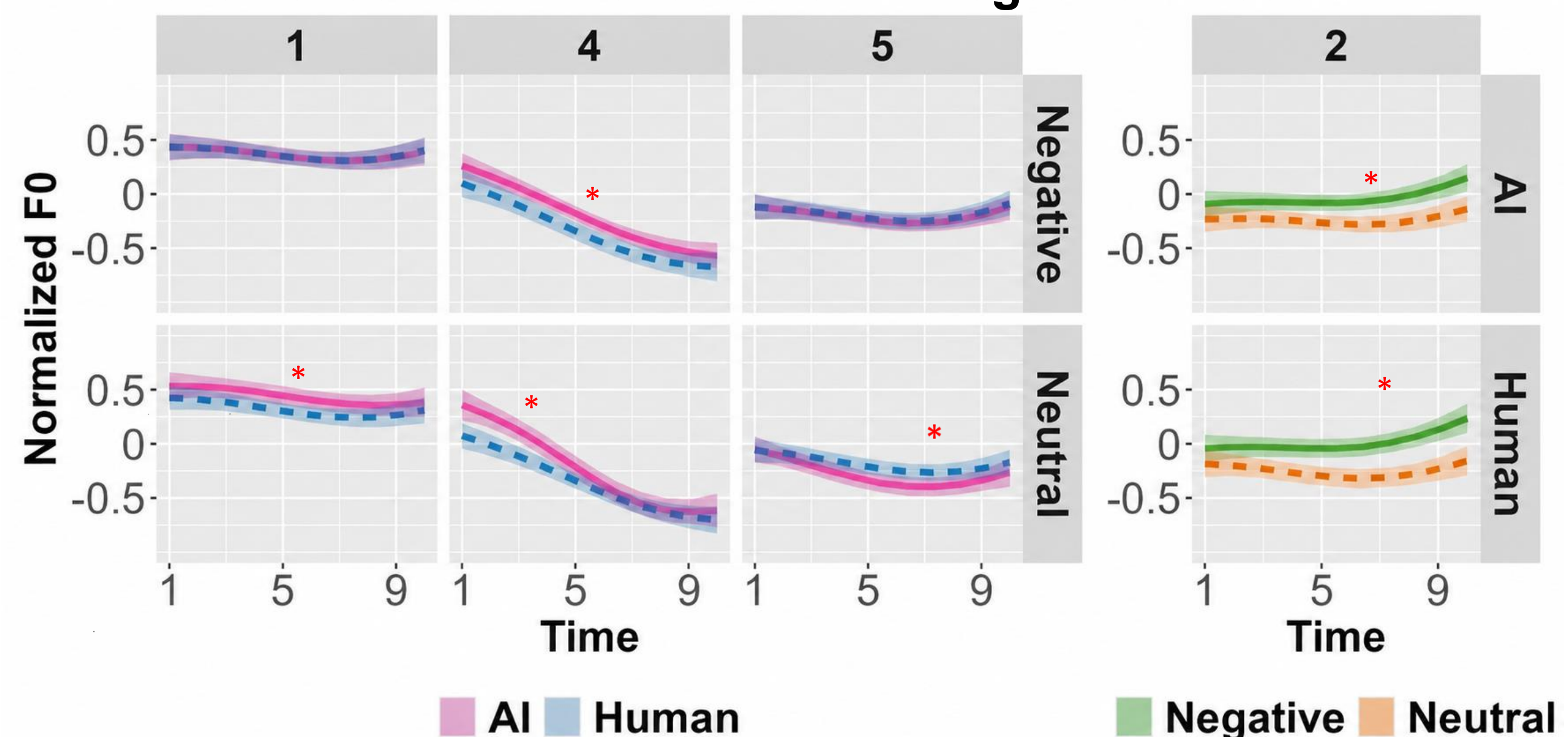


Figure 2: Smoothed curves of the tone contours produced contrasted by different interlocutors (left panels) and by different emotions (right panels).

Discussion

- Participants **adjusted specific tonal realizations** accordingly in response to difference in interaction between human and AI:
 - ✓ Interlocutor effect was **tone-specific** rather than a **general shift of F0 height**. It is possible that the participants were enlarging the pitch range or enhancing specific tonal features.
- **Negative** emotion elicited **more dispersed articulation** in **vowels** and a **specific tone**:
 - ✓ Consistent with accounts linking emotional intensity to **increased articulatory dispersion**, although the effect was modest.
- **AI** interaction elicited **shorter target word/syllable durations**:
 - ✓ This effect may be confounded by **differences in target-word selection** across interlocutors.
 - ✓ Our finding is more **robust** than previous studies in which the voice of the AI interlocutors was provided by existing AI assistants (e.g., Alexa), whose speech rate may **not** be **comparable** to the human interlocutors.
- **Limitations**:
 - ✓ Sample size: limited the power to detect the interlocutor and emotion effect
 - ✓ Rapid alternation of emotional context within brief dialogues may have limited phonetic modulation → ongoing work includes better designs

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